

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	14 July 2026
Team ID	_____
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	User Registration through Form
		User Login
		Password Recovery
FR-2	Dataset Management	Upload Electricity Consumption Dataset
		Validate Dataset
		Store Dataset in Database
FR-3	Electricity Data Analysis	Analyze Daily Consumption
		Analyze Monthly Consumption
		Analyze Yearly Consumption
FR-4	Reports & Visualization	Generate Graphs and Charts
		Generate Analysis Report
		Download Report (PDF/CSV)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Easy-to-use interface for viewing electricity consumption analysis.
NFR-2	Security	Secure login and protected user and dataset information.
NFR-3	Reliability	Provide accurate analysis with minimal system errors.
NFR-4	Performance	Process and visualize electricity data quickly.
NFR-5	Availability	System should be accessible whenever users need it.
NFR-6	Scalability	Support larger datasets and more users in the future.